

## Group NaMe :: DiGiTaL ThinKinG

Solved By: Abu Bakar Khalid, Abu Bakar Mehar(ABM),  
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lifetimemasty.blogspot.com](http://www.refinesol.com/lifetimemasty.blogspot.com)

1. For the sieve technique we solve the problem,  
**Recursively** (Handsouts Page 35 Line 4)  
mathematically  
precisely  
accurately
2. We do sorting to,  
keep elements in random positions  
keep the algorithm run in linear order  
keep the algorithm run in (log n) order  
**keep elements in increasing or decreasing order** (Handsouts Page 39 Line 6)
3. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,  
**divide-and-conquer** (handsouts page 34 )  
decrease and conquer  
greedy nature  
2-dimension Maxima
4. In Sieve Technique we don't know which item is of interest  
**True** (handsouts page 34)  
False
5. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,  
 $T(n)$   
 $T(n/2)$   
**log n** (handsouts page 37)  
 $n/2 + n/4$
6. Divide-and-conquer as breaking the problem into a small number of pivot  
Sieve  
**smaller sub problems** (handouts page 27)  
Selection
7. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_  
increasing order only  
decreasing order only  
**heap order** (handouts page 40)  
(log n) order

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8. Slow sorting algorithms run in,  $T(n^2)$   
 $T(n)$   
 $T(\log n)$   
 **$T(n \log n)$  (handsouts page 40)**
9. One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.  
**pointers (handsouts page 40)**  
constants  
variables  
functions
10. Sorting is one of the few problems where provable \_\_\_\_\_ bounds exists on how fast we can sort,  
upper  
**lower (handsouts page 39)**  
average  $\log n$   
2nd
11. For the sieve technique we solve the problem,  
mathematically  
precisely  
accurately  
**recursively (handsouts page 35)**
12. Sieve Technique can be applied to selection problem?  
**true (handsouts page 35)**  
false
13. How much time merge sort takes for an array of numbers? ( $n^2$ )  
 $T(n)$   
 $T(\log n)$   
 **$T(n \log n)$  (handsouts page 30)**
14. For the Sieve Technique we take time  
 **$T(nk)$  (handsouts page 34)**  
 $T(n/3) n^2$   
 $n/3$
15. Heaps can be stored in arrays without using any pointers; this is due to the \_\_\_\_\_ nature of the binary tree,  
**left-complete Repeat**  
right-complete tree nodes  
tree leaves

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16. How many elements do we eliminate in each time for the Analysis of Selection algorithm?

$n / 2$  elements

**$(n / 2) + n$  elements**

$n / 4$  elements  $2n$  elements

17. We do sorting to,

keep elements in random positions

keep the algorithm run in linear order

keep the algorithm run in  $(\log n)$  order

**keep elements in increasing or decreasing order ( page 39 Repeat)**

18. In which order we can sort?

increasing order only

decreasing order only

**increasing order or decreasing order Repeat**

both at the same time

19. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_

increasing order only

decreasing order only

**heap order (page 40)**

$(\log n)$  order

20. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

$T(n)$

$T(n / 2)$

**$\log n$  (page 37)**

$n / 2 + n / 4$

21. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_

increasing order only

decreasing order only

**heap order (page 40)**

$(\log n)$  order

22. How much time merge sort takes for an array of numbers?  $T(n^2)$

$T(n)$

$T(\log n)$

**$T(n \log n)$  (page 30)**

24. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent \_\_\_\_\_ series in the analysis,

linear

arithmetic

**geometric (page 37)**

exponent

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25. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of \_\_\_\_\_

**n items (page 34)**

phases

pointers

constant

26. A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order

**heap (page 40)**

binary tree

binary search tree

array

27. The sieve technique works in \_\_\_\_\_ as follows

**phases (page 34)**

numbers

integers

routines

29. For the heap sort, access to nodes involves simple \_\_\_\_\_ operations.

**arithmetic (page 41)**

binary

algebraic

logarithmic

30. The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in n

arithmetic

geometric

**linear (page 37)**

orthogonal

For the heap sort, access to nodes involves simple \_\_\_\_\_ operations. Select correct option:

**Arithmetic (page 41)**

binary

algebraic

logarithmic

In Sieve Technique we do not know which item is of interest

**True (page 34)**

False

How much time merge sort takes for an array of numbers?

$T(n^2)$   $T(n)$

$T(\log n)$

**$T(n \log n)$  (page 30)**

For the heap sort we store the tree nodes in

**level-order traversal (page 40)**

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in-order traversal pre-order traversal  
post-order traversal  
One of the clever aspects of heaps is that they can be stored in arrays without using any

\_\_\_\_\_  
**Pointers (page 40)**  
constants  
variables  
functions

Sorting is one of the few problems where provable \_\_\_\_\_ bounds exist on how fast we can sort,

upper  
**lower (page 39)**  
average log n

A heap is a left-complete binary tree that conforms to the \_\_\_\_\_  
Select correct option:

increasing order only  
decreasing order only  
**heap order** **Repeat**  
(log n) order

In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

$T(n)$   
 $T(n/2)$   
**log n** **Repeat**  
 $n/2 + n/4$

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

**divide-and-conquer (page 34)**  
decrease and conquer  
greedy nature  
2-dimension Maxima

The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in n,  
Select correct option:

arithmetic  
geometric  
**linear** **Repeat**  
orthogonal

The sieve technique works in \_\_\_\_\_ as follows

**Phases** **Repeat**  
numbers  
integers  
routines

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Sorting is one of the few problems where provable \_\_\_\_\_ bounds exist on how fast we can sort,

upper

**lower**      **Repeat**

average

$\log n$

In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

Select correct option:

$T(n)$

$T(n/2)$

**$\log n$**       **Repeat**

$n/2 + n/4$

For the Sieve Technique we take time

**$T(nk)$**       **Repeat**

$T(n/3)$

$n^2$

$n/3$

A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order Select correct option:

**heap**      **Repeat**

binary tree

binary search tree array

For the heap sort we store the tree nodes in

**level-order traversal**      **Repeat**

in-order traversal

pre-order traversal

post-order traversal

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent \_\_\_\_\_ series in the analysis,

linear

arithmetic

**geometric**      **Repeat**

exponent

One of the clever aspects of heaps is that they can be stored in arrays without using any

\_\_\_\_\_.  
**Pointers**      **Repeat**

constants

variables

functions

Analysis of Selection algorithm ends up with,

**$T(n)$**       **(page 37)**

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$T(1 / 1 + n)$   
 $T(n / 2)$   
 $T((n / 2) + n)$

The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in  $n$   
arithmetic  
geometric

**Linear**      **Repeat**  
orthogonal

**Question No. 1 Marks : 1**

**Consider the following pairs of functions**

**I.  $f(x) = x^2 + 3x + 7$   $g(x) = x^2 + 10$**

**II  $f(x) = x^2 \log(x)$   $g(x) = x^3$**

**III  $f(x) = x^4 + \log(3x^8 + 7)$   $g(x) = (x^2 + 17x + 3)^2$**

**Which of the pairs of functions  $f$  and  $g$  are asymptotic?**

**Only I**      **Sure**  
Only II  
Both I and III  
None of the above

**Question No. 2 Marks : 1**

**Execution of the following code fragment**

```
int Idx;  
for (Idx = 0; Idx < N; Idx++)  
{  
    cout << A[Idx] << endl;  
}
```

**is best described as being**

**$O(N)$**       **Sure**  
 $O(N^2)$   
 $O(\log N)$   
 $O(N \log N)$

**Question No. 3 Marks : 1**

**If algorithm  $A$  has running time  $7n^2 + 2n + 3$  and algorithm  $B$  has running time  $2n^2$ , then**

**Both have same asymptotic time complexity**      **Sure**  
 $A$  is asymptotically greater  
 $B$  is asymptotically greater  
None of others

**Question No. 4 Marks : 1**

**Which of the following sorting algorithms is stable?**

**(i) Merge sort,**

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- (ii) Quick sort,
- (iii) Heap sort,
- (iv) Counting Sort.

Only i

Only ii

Both i and ii

Both iii and iv **Ref (Merge sort and counting sort are stable algorithms)**

**Question No. 1 Marks : 5**

**The following subroutine computes for a given number N.**

**compute(N)**

**{**

**If (N==1)**

**return 2**

**else**

**return compute(N - 1) \* compute(N - 1)**

**}**

**What category of algorithmic solutions best characterizes the approach taken in this subroutine (algorithm)?**

**Search and traversal**

**(Not Sure)**

**Divide-and-conquer**

**Greedy algorithm**

**Dynamic Programming**

**Question No. 1**

**Assume that a given algorithm has a runtime C that depends on the size N of its input**

**according to the following two formulas:**

**0 1**

**()**

**(1) 2**

**if N**

**C N**

**C N if N**

**=**

**= - + > 2**

**Which of the following functions C(N) describes the runtime of the algorithm?**

**C(N) = N - 1**

**C(N) = (N - 1)2**

**C(N) = log<sub>2</sub> N**

**(Not Sure)**

**C(N) = 2(N - 1)**

**Question No. 7**

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Let us say we have an algorithm that carries out  $N^2$  operations for an input of size  $N$ . Let us say that a computer takes 1 microsecond ( $1/1000000$  second) to carry out one operation. How long does the algorithm run for an input of size 3000?

90 seconds

**9 seconds** (Sure)

0.9 seconds

0.09 seconds

**Question No. 8**

**Consider the following polynomial**

$akn^k + ak-1nk-1 + \dots + a_0$ .

**What is the Big -O representation of the above polynomial?**

$O(kn)$

**$O(nk)$**  (Sure)

$O(nk+1)$

None of the above

**1. For the Sieve Technique we take time**

>  **$T(nk)$**  (page 34)

>  $T(n/3)$

>  $n^2 > n/3$

**2. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of \_\_\_\_\_**

Select correct option:

>  **$n$  items** (page 34)

> phases

> pointers

> constant

**3. \_\_\_\_\_ graphical representation of al-**

gorithm.

> Asymptotic

> **Flowchart** (rep)

**4. who invented the quick sort**

**Hoare (1961, 1962)** ([click here](#))

**5. function is given like  $4n^4 + 5n^3 + n$  what is the run time of**

**this**

>  **$\theta(n^4)$**  ([click here](#))

>  $\theta(n^3)$

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$\Theta(4n^4 + 5n^3)$

$\Theta(4n^4 + 5n^3)$

**6. main elements to a divide-and-conquer**

>Divide

>conquer

>combine

**ALL of above (page 27)**

**8.  $T(n) = \begin{cases} 4 & \text{if } n=1, \\ \text{otherwise } T(n/5) + 3n \end{cases}$  what is the answer if  $n=5$**

**Answer is 79 (not sure)**

**8. Merge sort is a stable algorithm but not an in-place algorithm.**

**>True (page 54)**

>false

**9. Counting sort the numbers to be sorted are in the range 1 to k where k is small.**

**>True (page 57)**

>False

**1 Total time for heapify is:**

$O(\log^2 n)$

$O(n \log n)$

$O(n^2 \log n)$

**$O(\log n)$  (page 43)**

**2 Solve the recurrence using iteration method and also find time complexity ( $\Theta$  notation)**

$T(n) = C + O(1) + T(n-1)$

$T(1) = 1$  and  $C$  is a constant.

**3 If an algorithm has a complexity of  $\log_2 n + n \log_2 n + n$ . we could say that it has complexity**

$O(n)$

$O(n \log_2 n)$

$O(3)$

$O(\log_2 (\log_2 n))$

**$O(\log_2 n)$  (Not sure > [click here](#))**

**4. In RAM model instructions are executed**

One after another

Parallel

Concurrent

**Random (page 10)**

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5. In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

**Convergent geometric series** (page 37)

Divergent geometric series

None of these

6. Due to left-complete nature of binary tree, heaps can be stored in

Link list

Structure

**Array** (page 40)

None of above

**Question No: 1 ( Marks: 1 ) - Please choose one**

**Random access machine or RAM is a/an**

☐ Machine build by Al-Khwarizmi

☐ Mechanical machine

☐ Electronics machine

☐ **Mathematical model** ref (Page 10)

**Question No: 2 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ is a graphical representation of an algorithm

☐ S notation

☐ Q notation

☐ **Flowchart** ref (<http://www.edrawsoft.com/flowchart.php>)

☐ Asymptotic notation

**Question No: 3 ( Marks: 1 ) - Please choose one**

A RAM is an idealized machine with \_\_\_\_\_ random-access memory.

☐ 256MB

☐ 512MB

☐ **an infinitely large** ref (Page 10)

☐ 100GB

**Question No: 4 ( Marks: 1 ) - Please choose one**

**What type of instructions Random Access Machine (RAM) can execute? Choose best answer**

☐ Algebraic and logic

☐ Geometric and arithmetic

☐ **Arithmetic and logic** ref (Page 10)

☐ Parallel and recursive

**Question No: 5 ( Marks: 1 ) - Please choose one**

**What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?**

☐  $\frac{n}{2}$

☐  $n^2$

☐ n

ref(Page 12) there are two loops

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$n^8$

**Question No: 6 ( Marks: 1 ) - Please choose one**

**What is the solution to the recurrence  $T(n) = T(n/2) + n$  .**

- ☐  $O(\log n)$
- ☒  $O(n)$  ref(ref book page # 33)
- ☐  $O(n \log n)$
- ☐  $O(n^2)$

**Question No: 7 ( Marks: 1 ) - Please choose one**

**Consider the following code:**

```
For(j=1; j<n; j++)
    For(k=1; k<15; k++)
        For(l=5; l<n; l++)
            {
                Do_something_constant();
            }
```

**What is the order of execution for this code.**

- ☐  $O(n)$
- ☒  $O(n^3)$  ref (there are 3 nested for loops so it's order is  $O(n^3)$ )
- ☐  $O(n^2 \log n)$
- ☐  $O(n^2)$

**Question No: 8 ( Marks: 1 ) - Please choose one**

**Consider the following Algorithm:**

```
Factorial (n){
if (n=1)
return 1
else
return (n * Factorial(n-1))
```

**Recurrence for the following algorithm is:**

- ☐  $T(n) = T(n-1) + 1$
- ☐  $T(n) = nT(n-1) + 1$
- ☐  $T(n) = T(n-1) + n$
- ☒  $T(n) = T(n-1) + 1$  (Not Sure)

**Question No: 9 ( Marks: 1 ) - Please choose one**

**What is the total time to heapify?**

- ☐  $(\log n)$  ref (Page # 43)
- ☐  $(n \log n)$
- ☐  $(n^2 \log n)$
- ☐  $(\log^2 n)$

**Question No: 10 ( Marks: 1 ) - Please choose one**

**When we call heapify then at each level the comparison performed takes time**

**It will take  $\theta(1)$**  ref (Page # 43)

Time will vary according to the nature of input data

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It can not be predicted

It will take  $\theta(\log n)$

**Question No: 11 ( Marks: 1 ) - Please choose one**

**In Quick sort, we don't have the control over the sizes of recursive calls**

☐ True ref (page # 49)

☐ False

☐ Less information to decide

☐ Either true or false

**Question No: 12 ( Marks: 1 ) - Please choose one**

**Is it possible to sort without making comparisons?**

☐ Yes ref (page # 57)

☐ No

**Question No: 13 ( Marks: 1 ) - Please choose one**

**If there are  $n^2$  entries in edit distance matrix then the total running time is**

(1)

$(n^2)$

$(n)$

$(n \log n)$

**Question No: 14 ( Marks: 1 ) - Please choose one**

**For Chain Matrix Multiplication we can not use divide and conquer approach because,**

**We do not know the optimum k** ref (page # 86)

We use divide and conquer for sorting only

We can easily perform it in linear time

Size of data is not given

**Question No: 15 ( Marks: 1 ) - Please choose one**

**The Knapsack problem belongs to the domain of \_\_\_\_\_ problems.**

**Optimization** ref (page # 91)

NP Complete

Linear Solution

Sorting

**Q 1**

Total time for heapify is:

$(\log^2 n)$

$(n \log n)$

$(n^2 \log n)$

**$(\log n)$  (page 43) repeat**

Q2: \_

If an algorithm has a complexity of  $\log^2 n + n \log^2 n + n$ . we could say that it has complexity

1)  $O(n)$

**2)  $O(n \log^2 n)$  (page 106)**

3)  $O(3)$

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4)  $O(\log^2(\log 2n))$  5)  $O(\log 2n)$

Q 4

Due to left-complete nature of binary tree, heaps can be stored in Link list Structure

**Array (page 40) repeat**

None of above

Q5

In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

**Convergent geometric series (page 37)**

Divergent geometric series None of these

**Q#6**

In RAM model instructions are executed

**One after another (page 10)**

Parallel

Concurrent

Random

Qus 7

Consider the following pairs of functions

I.  $f(x) = x + 3x + 7$   $g(x) = x + 1022$

II  $f(x) = x^2 \log(x)$   $g(x) = x^3$

III  $f(x) = x + \log(3x + 7)$   $g(x) = (x^4 2 + 17x + 3)^2$

Which of the pairs of functions  $f$  and  $g$  are asymptotic?

**Only I not sure**

Only II

Both I and III

None of the above

**Q8**

Execution of the following code fragment

```
int Idx;  
for (Idx = 0; Idx < N; Idx++)
```

```
{  
    cout << A[Idx] << endl;  
}
```

is best described as being

**$O(N)$  sure**

$O(N^2)$

$O(\log N)$

$O(N \log N)$

Q9

If algorithm  $A$  has running time  $7n^2 + 2n + 3$  and algorithm  $B$  has running time  $2n^2$ , then

**Both have same asymptotic time complexity (see page 24)**

$A$  is asymptotically greater

$B$  is asymptotically greater

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None of others

**Question No. 10**

Which of the following sorting algorithms is stable?

Merge sort,

Quick sort,

Heap sort,

Counting Sort.

Only i

Only ii

Both i and ii

Both iii and iv

**“Counting sort and merge sort are both stable algorithms “**

Q11

Consider the following recurrence relation

$T(n) = 0$  if  $n = 0$   $T(n) = ?$

$6(T(n-1))$  otherwise

Then  $T(2)$  is

**? 36 not sure**

72

12

None of the above

Q12

Let us say we have an algorithm that carries out  $N^2$  operations for an input of size  $N$ . Let us say that a computer takes 1 microsecond ( $1/1000000$  second) to carry out one operation. How long does the algorithm run for an input of size 3000?

? 90 seconds ? 9 seconds

? 0.9seconds

**? 0.09 seconds**

Q13

The appropriate big  $\theta$  classification of the given function.  $f(n) = 4n^2 + 97n + 1000$  is

? (n)

? (2n)

**? (n<sup>2</sup>)**

? (n<sup>2</sup> log n)

Q14

Consider the following polynomial

$a_k n^k + a_{k-1} n^{k-1} + \dots + a_0$ .

What is the Big  $\theta$  representation of the above polynomial?

$O(kn)$

**$O(nk)$**

$O(nk+1)$

None of the above

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Q15

Consider the C++ program segment given below: for ( i=0 ; i < n; i++) {  
for (j = 1,sum=a[0]; j <= i; j++) Sum+=a[j];  
Cout<< "The sum for sub-array is<< sum;  
}

The running time of the above algorithm is

n.

$n \log n$ .

$\log n$ .

**n<sup>2</sup>.**

Q16

Consider the following pairs of functions

I.  $f(x) = (x^3 + x^2 + x + 1)^4$   $g(x) = (x^4 + x^3 + x^2 + x + 1)^3$

II  $f(x) = 22x$   $g(x) = 2x^2$

III  $f(x) = 2x + 3$   $g(x) = 2x + 7$

IV  $f(x) = \log(x^2 + 1)$   $g(x) = \log(x)$

Which of the pairs of functions f and g are asymptotic?

Only I

Only III

Both I and II

**Both III and IV**

**Question No. 5**

Dynamic programming uses a top-down approach.

**? True [click here](#)**

? False

**Question No. 6**

The edit distance between *FOOD* and *MONEY* is

**At most four (page 76)**

At least four

Exact four

**Question No. 7**

Consider the following recurrence relation

4 if  $n = 1$   $T(n) = ?$

$T(n) + 2$

$n$  if  $n$  is divisible by 5 ( $n/5$ ) 3

Then  $T(5)$  is

**25**

**Not sure**

75

79

None of the above

**Question No. 10**

Execution of the following code fragment int i = N;

while (i > 0)

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```
{  
int Sum = 0; int j;  
for (j = 0; j < N; j++) Sum++;  
cout << Sum << endl; i--;  
}
```

is best described as being

**O(N) Sure**

O(N<sup>2</sup>)

O(log N) O(N

log N) **Question**

**No. 11**

It is impossible to design a sorting algorithm based on comparison of keys whose worst case run time is in  $O(n)$ .

? True

? **False (page 55)**

**Question No. 12**

Consider the following recurrence relation

5 if  $n = 1$

$T(n) = ? + \text{if } n \text{ is even}$

? 2 ( / 2 ) 3 Then  $T(8)$  is

61

29

13

**None of the above Not sure**

**Question No. 13**

The merge sort algorithm involves the following steps.

Recursively sort the 1st and 2nd halves separately

Merge the two-sorted halves into a sorted group.

If the number of items to sort is 0 or 1, return.

Which is the correct order of instructions in merge sort algorithm?

? (i),(ii),(iii)

? (ii),(iii),(i)

? (iii),(ii),(i)

? **(iii),(i),(ii) (page 27)**

**qus14**

iterfib achieve an exponential speedup over the original recursive algorithm

**True (page 75)**

False

Q What type of instructions Random access machine can execute?

Choose best answer.

Geometric and arithmetic

Algebraic and logic

**Arithmetic and logic (Page 10)**

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Parallel and recursive

Q Due to left complete nature of binary tree, the heap can be stored in

- Arrays (page 40)
- Structures
- Link Lis
- Stack

Q What type of instructions Random Access Machine (RAM) can execute? Choose best

answer

Algebraic and logic

Geometric and arithmetic

**Arithmetic and logic rep(Page 10)**

Parallel and recursive

Q For Chain Matrix Multiplication we can not use divide and conquer approach because,

**We do not know the optimum k (page 86)**

We use divide and conquer for sorting only

We can easily perform it in linear time

Size of data is not given

Q knapsack problem is called a “0-1” problem, because  
????????????????????

**Each item must be entirely accepted or rejected (page 92)**

????????????????????

????????????????????

Q word Algorithm comes from the name of the muslim author:

**Abu Ja'far Mohammad ibn Musa al-Khwarizmi. (page 7)**

Q al-Khwarizmi's work was written in a book titled

**al Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabalah (page 7)**

Q What is the total time to heapify?

- **$O(\log n)$  (page 43)**
- $O(n \log n)$
- $O(n^2 \log n)$
- $O(\log^2 n)$

**1. For the sieve technique we solve the problem, recursively (page 34)**

mathematically

precisely

accurately

**2. We do sorting to,**

keep elements in random positions

keep the algorithm run in linear order

keep the algorithm run in  $(\log n)$  order

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- keep elements in increasing or decreasing order (page 39)
3. the reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,  
divide-and-conquer (Page 34)  
decrease and conquer  
greedy nature  
2-dimension Maxima
4. In Sieve Technique we do not know which item is of interest  
True (page 34)  
False
5. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,  
 $T(n)$   
 $T(n / 2)$   
 $\log n$  (page 37)  
 $n / 2 + n / 4$
6. Divide-and-conquer as breaking the problem into a small number of  
pivot  
Sieve  
Smaller subproblems (page 34)  
Selection
7. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_  
increasing order only  
decreasing order only  
heap order (page 40)  
 $(\log n)$  order
8. Slow sorting algorithms run in,  
 $T(n^2)$  (page 39)  
 $T(n)$   
 $T(\log n)$   
 $T(n \log n)$
9. One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.  
pointers (page 40)  
constants  
variables  
functions
10. Sorting is one of the few problems where provable \_\_\_\_\_ bounds exists on how fast we can sort,  
upper  
lower (page 39)  
average  
 $\log n$

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2nd

- 11. For the sieve technique we solve the problem,**  
mathematically  
precisely  
accurately  
**recursively (page 34) Repeat**
- 12. Sieve Technique can be applied to selection problem?**  
true (page 34)  
false
- 13. How much time merge sort takes for an array of numbers?**  
( $n^2$ )  
 $T(n)$   
 $T(\log n)$   
 **$T(n \log n)$  (page 40)**
- 14. For the Sieve Technique we take time**  
 **$T(nk)$  (page 34)**  
 $T(n / 3)$   
 $n^2$   
 $n/3$
- 15. Heaps can be stored in arrays without using any pointers; this is due to the \_\_\_\_\_ nature of the binary tree,**  
**left-complete (page 40)**  
right-complete  
tree nodes  
tree leaves
- 16. How many elements do we eliminate in each time for the Analysis of Selection algorithm?**  
 **$n / 2$  elements (page 36)**  
 $(n / 2) + n$  elements  
 $n / 4$  elements  
 $2n$  elements
- 17. We do sorting to,**  
keep elements in random positions  
keep the algorithm run in linear order  
keep the algorithm run in  $(\log n)$  order  
**keep elements in increasing or decreasing order (page 39) Repeat**
- 18. In which order we can sort?**  
increasing order only  
decreasing order only  
**increasing order or decreasing order**  
both at the same time
- 19. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_**  
increasing order only

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decreasing order only  
**heap order (page 40)**  
(log n) order

**20. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,**

$T(n)$   
 $T(n / 2)$   
**log n (page 37) Repeat**  
 $n / 2 + n / 4$

**21. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_**

increasing order only  
decreasing order only  
**heap order (page 40) Repeat**  
(log n) order

**22. How much time merge sort takes for an array of numbers?**

$T(n^2)$   
 $T(n)$   
 $T(\log n)$   
 **$T(n \log n)$  (page 40) repeat**

**23. One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.**

**pointers (page 40) Repeat**  
constants  
variables  
functions

**24. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent \_\_\_\_\_ series in the analysis,**

linear  
arithmetic  
**geometric (page 37)**  
exponent

**25. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of \_\_\_\_\_**

**n items (page 34)**  
phases  
pointers  
constant

**26. A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order**

**heap** **ref (page # 40)**  
binary tree  
binary search tree  
array

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27. The sieve technique works in \_\_\_\_\_ as follows  
phases ref (page # 34)

numbers  
integers  
routines

28. For the sieve technique we solve the problem,  
recursively ref (page #35)

mathematically  
precisely  
accurately

29. For the heap sort, access to nodes involves simple \_\_\_\_\_  
operations.

arithmetic ref (page # 40)  
binary  
algebraic  
logarithmic

30. The analysis of Selection algorithm shows the total running time is  
indeed \_\_\_\_\_ in  $n$ ,

arithmetic  
geometric  
linear ref (page # 37)  
orthogonal

For the sieve technique we solve the problem,  
Select correct option:

recursively ref (page # 35)  
mathematically  
precisely  
accurately

For the heap sort, access to nodes involves simple \_\_\_\_\_  
operations.

Select correct option:

arithmetic ref (page # 40)  
binary  
algebraic  
logarithmic

Quiz Start Time: 07:39 AM

Time Left 76

sec(s)

We do sorting to,

Select correct option:

keep elements in random positions  
keep the algorithm run in linear order  
keep the algorithm run in  $(\log n)$  order

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keep elements in increasing or decreasing order  
One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.

Select correct option:

- pointers** ref (page # 40)
- constants
- variables

A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order

Select correct option:

- heap** rep
- binary tree
- binary search tree
- array

The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in n,

Select correct option:

- arithmetic** rep
- geometric
- linear
- orthogonal

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of \_\_\_\_\_

Select correct option:

- n items** ref (page # 34)
- phases
- pointers
- constant

Divide-and-conquer as breaking the problem into a small number of

Select correct option:

- pivot
- Sieve
- smaller sub problems** ref (page # 34)
- Selection

In Sieve Technique we do not know which item is of interest

Select correct option:

- True** ref (page # 34)
- False

How much time merge sort takes for an array of numbers?

Select correct option:

- $T(n^2)$
- $T(n)$
- $T(\log n)$

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$T(n \log n)$  ref (page # 40)

For the heap sort we store the tree nodes in

Select correct option:

level-order traversal ref (page # 40)

in-order traversal

pre-order traversal

post-order traversal

1. For the heap sort we store the tree nodes in

level-order traversal (page 40)

in-order traversal

pre-order traversal

post-order traversal

2. One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.

pointers (page 40)

constants

variables

functions

3. Sorting is one of the few problems where provable \_\_\_\_\_ bounds exists on how fast we can sort,

upper

lower (page 39)

average

$\log n$

4. A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order

heap (page 40)

binary tree

binary search tree

array

5. Sieve Technique applies to problems where we are interested in finding a Single item from a larger set of \_\_\_\_\_

$n$  items (page 34)

phases

pointers

constant

6. How much time merge sort takes for an array of numbers?

$T(n^2)$

$T(n)$

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$T(\log n)$

**$T(n \log n)$  (page40)**

7. A heap is a left-complete binary tree that conforms to the \_\_\_\_\_  
increasing order only  
decreasing order only  
**heap order (page40)**  
(log n) order

8. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

$T(n)$

$T(n / 2)$

**log n (page 37)**

$n / 2 + n / 4$

9. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

**divide-and-conquer (page34)**

decrease and conquer

greedy nature

2-dimension Maxima

10. The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in n,

arithmetic

geometric

**linear (page37)**

orthogonal

1. The sieve technique works in \_\_\_\_\_ as follows

**phases (page34)**

numbers

integers

routines

2. Sorting is one of the few problems where provable \_\_\_\_\_ bounds exists on how fast we can sort,

Select correct option:

upper

**lower (rep 39)**

average

log n

3. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

Select correct option:

$T(n)$

**log n (rep 37)**

$n / 2 + n / 4$

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**4. For the Sieve Technique we take time**

Select correct option:

**$T(nk)$  (page 34)**

$T(n / 3)$

$n^2$

$n/3$

**5. A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order**

Select correct option:

**heap (page 40)**

binary tree

binary search tree

array

**6. For the heap sort we store the tree nodes in**

Select correct option:

**level-order traversal (rep 40)**

in-order traversal

pre-order traversal

post-order traversal

**7. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent \_\_\_\_\_ series in the analysis,**

Select correct option:

linear

arithmetic

**geometric (rep 37)**

exponent

**8. One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.**

Select correct option:

**pointers (rep 40)**

constants

variables

functions

**Sorting is one of the few problems where provable \_\_\_\_\_ bounds exist on how fast we can sort,**

Select correct option:

upper

lower (page 39)

average

$\log n$

**In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,**

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Select correct option:

$T(n)$  (page 39)

$\log n$

$n/2 + n/4$

**For the Sieve Technique we take time**

Select correct option:

$T(nk)$  (page 34)

$T(n/3)$

$n^2$

$n/3$

**A (an) \_\_\_\_\_ is a left-complete binary tree that conforms to the heap order**

Select correct option:

heap (page 40)

binary tree

binary search tree

array

**For the heap sort we store the tree nodes in**

Select correct option:

level-order traversal (page 40)

in-order traversal

pre-order traversal

post-order traversal

**In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent \_\_\_\_\_ series in the analysis,**

Select correct option:

linear

arithmetic

geometric (page 37)

exponent

**One of the clever aspects of heaps is that they can be stored in arrays without using any \_\_\_\_\_.**

Select correct option:

pointers (page 40)

constants

variables

functions

**Analysis of Selection algorithm ends up with,**

Select correct option:

$T(n)$  repeat

$T(1/1 + n)$

$T(n/2)$

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$$T((n/2) + n)$$

The analysis of Selection algorithm shows the total running time is indeed \_\_\_\_\_ in  $n$ ,

Select correct option:

arithmetic

geometric

linear (page 39)

orthogonal